CCCA Meta-Atlas · Cell-State Manifold Analysis

Wu et al. 2020 · Acute Myeloid Leukemia · Microwell-seq · 96,881 malignant cells (36 samples)

GO / NO-GO GATES

Gate 1 · integration + continuity — GO

Harmony θ=10 on `sample` (iLISI 9.37 of 23.63 ceiling — climbs to θ=10, drops at θ=20). After removing 20 cells in 17 detached specks, all 96,881 cells form ONE connected component: samples intermix per-neighbourhood on a single continuous manifold. Judged on the regenerated embedding, not a scalar alone.

Gate 2 · embryonic module geography — GO (non-branching)

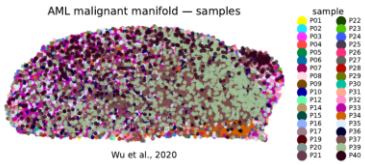
Single Gast/pSom-high → Crest-high developmental axis. Organized Gast↔pSom cross-repression (smoothed r = −0.68); Crest opposes Gast (r = −0.67), co-varies weakly with pSom (r = +0.11) and rises to the terminus (Crest-high decile at pseudotime 0.82; Spearman ρ = +0.43). No Gast/pSom co-expression hub and no opposing Gast arm → non-branching, not a bifurcation. Crest terminal, not on the Gast arm (no inversion flag).

Gate 3 · hub cross-repression — NA

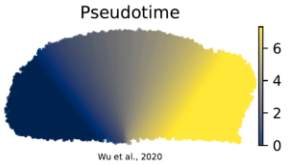
Branching-only. The manifold does not bifurcate — the diagnostic tree fit shows no Gast/pSom co-expression hub and no opposing Gast arm (both non-root tips are pSom-high / Crest-low), and the committed curve is fork-free. There is no hub population to test, so Gate 3 does not apply.

STEP FIGURES

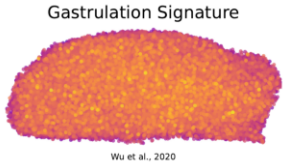
01 · Samples (integration)



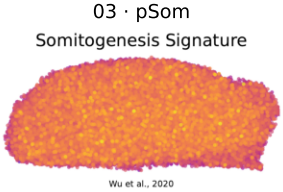
02 · Pseudotime



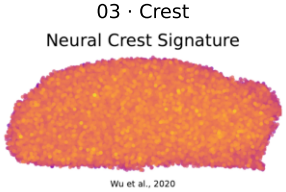
03 · Gast



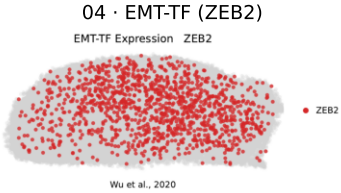
03 · pSom



03 · Crest



04 · EMT-TF (ZEB2)



PARAMETER CHOICES & RATIONALE

- **Min-cell threshold:** ≥100 (floor) → 96,901 cells, 36/38 samples
  Per-sample counts smooth down to 417, then two specks (98, 31) fall below the floor; the top gap is not a dominant spillover cliff, so the ≥100 viability floor applies. Dropped P18, P13.
- **Integration:** Harmony θ=10 · n_pcs=20 · n_neighbors=30 · spread=1.3
  θ swept 1/2/5/10/20; iLISI plateaus at θ=10 then falls at θ=20 — strongest integration without collapse. 443 over-dispersed genes.
- **Island removal:** −20 cells (17 specks) → 96,881
  UMAP radius graph at 6× median NN distance; largest connected component retained.
- **Geometry:** Non-branching CURVE (Linear-pSom/Crest)
  Tree fit → no Gast/pSom co-expression hub; curve N=20 fold-free (self-approach 2.90); continuity |corr(t, DC1)| = 0.91 on an independent diffmap. Raw module means: pSom +0.007 > Gast +0.002 > Crest −0.004.
- **Root:** Curve tip at the Gast/pSom-high, Crest-low pole
  Rooted from developmental order (Gast/pSom EARLY, Crest LATE); pseudotime not proliferation-driven (ρ vs cycling = −0.07; single low-cycling body, cycling fraction 0.001).
- **EMT-TF readout:** ZEB2 (sole detectable canonical EMT-TF, 5.1%)
  Broad panel screened (Snail/Zeb/Twist/FOXC/PRRX); FOXC1/2, PRRX1/2, SNAI2, TWIST1 absent from the Microwell-seq gene panel and SNAI1/ZEB1/TWIST2 below the 3% detection floor. Shown as a categorical overlay.

VERDICT

The embryonic-program pattern reproduces in a NON-BRANCHING form: one continuous, sample-integrated AML manifold carrying a single Gastrulation/Somitogenesis-high → Neural-crest-high developmental axis with organized Gast↔pSom cross-repression (r = −0.68) and a terminal Crest gradient. No bifurcation, so Gate 3 is NA. Gates 1–2 GO; geometry Linear-pSom/Crest.